Code flow:-

1. Data cleaning

1. Import Pandas, Numpy and Seaborn libraries.
2. Load the dataset
3. Check and fill missing values using median
4. Convert categorical values into numerical values using get_dummies

Dataset:- Titanic - 891 rows, 6 columns

5. Cleaned dataset - display

2. Train and Test data:-

1. Import the requried libraries
2. Load the Iris dataset
3. Split the dataset into training & testing data
4. Apply KNN algorithm
5. Calcaute and display the accuracy

Data Set:- Iris - 150 rows, 4 features, 3 classes

3. Linear regression:

1. Import Pandas, Numpy and scikit-learn
2. Load the Auto MPG dataset
3. Select input and output features
4. Train th
5. Predict

Data set:-

4. KNN a
1. Import
2. Load
3. Separa
4. Split
5. Apply
accur

Dataset

5. K-HC
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Data

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1. Im
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4. Train the linear regression model
5. Predict mileage and display the result.

Data set :- Auto MPG - 398 rows

4. KNN and SVM classification:

1. Import the required libraries
2. Load the Social Network Ads dataset.
3. Separate input and output features
4. Split data into training and testing sets
5. Apply KNN and SVM algorithms and display accuracy

Dataset :- Social Network Ads - 400 rows

5. K-Means clustering :-

1. Import the required libraries
2. Load the mall customers data set
3. select the required features
4. Apply Hircarchical clustering
5. Draw and display the dendrogram

Dataset : mall customer - 200 rows

6. Hierarchical clustering :

1. Import the required libraries
2. Load the Breast Cancer dataset

3. select the required features
4. Apply Hircarchical clustering
5. Draw and display the dendogram

Data Set: Breast Canter mall customers - 200 rows

7. Random Forest :-

1. Import the requried libraries
2. Load the Breast Canter dataset
3. Split the data into training and testing set
4. Apply the random forest algorithm
5. Predict the result and Calcaute accuracy

Dataset:- Breast Canter - 569 rows, 30 features

8. Boosting

1. Import the requried libraries
2. Load the heart disease dataset
3. Seperate input and output features
4. Train the Boosting model
5. Display the learned result & accuracy

Data Set:- Heart Disease - 303 rows, 13 features

9- Q-learning:-

- Import the Gymnasium and Num... libraries

2. Create the forzen lake game enviorment
3. Initialize the Q-table
4. Train the agent using rewards and penalites
5. Disply the learned result.

Data set:- forzen lake - 4x4 game enviorment

10. Neural Network:

1. Import Tensor flow and scikit-learn libraries
2. Load the Hand written digits Data sets
3. Split the data into training and testing sets
4. Build and train the Neural Network model
5. Predict the hand written digits and digita accuracy

Data set: Digits - 1797 images, 8x8 pixels